Formula 1 Podium Prediction

Milestone 1 analytical report

1. Overview

The task is to predict whether a driver finishes on the podium (`positionOrder <= 3`) using only information available after qualifying and before the race starts. One row is one driver in one race. Results, points, laps, status, pit stops, lap times and the standings published for the same race are not used as features.

The selected model is a random forest. On the unseen test seasons (2022–2024) it reaches PR-AUC 0.715, ROC-AUC 0.938 and F1 0.683, against PR-AUC 0.647 for a simple grid rule and 0.150 for chance.

2. Data cleaning and EDA

- **Sentinels and types.** `\N` was replaced with missing values; dates, numbers and lap-time strings were typed (0 conversion failures). In `constructor_results.status`, `\N` means “not disqualified” and was coded as such.
- **Keys.** 39 of 42 key checks passed before cleaning and 42 of 43 after. The lap-time key was repaired: 1,710 exact copies dropped and 269 conflicting copies resolved and logged.
- **Duplicates.** 85 driver-race pairs (176 rows, 1950–1978, mostly shared drives) were collapsed by keeping the best-classified stint. The alternative rule (most laps) keeps the same row in all 85 cases. Result: 27,499 rows, one per (raceId, driverId).
- **Coverage.** Qualifying starts in 1994 and is complete only from 2003, so modeling uses 2003–2024. A grid of 0 means “no grid slot” and never produces a podium.

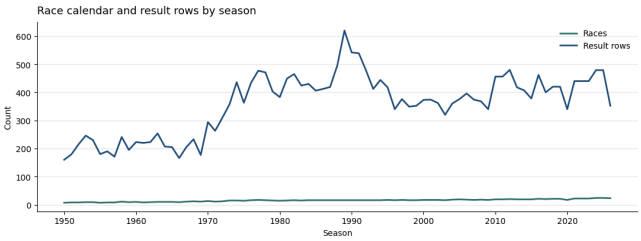


Figure 1. Races and result rows by season.

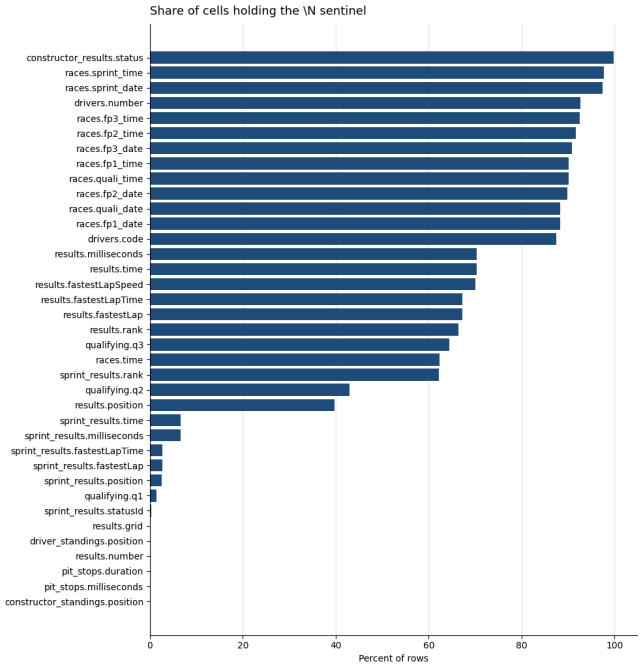


Figure 2. Share of `\N` sentinels by column.

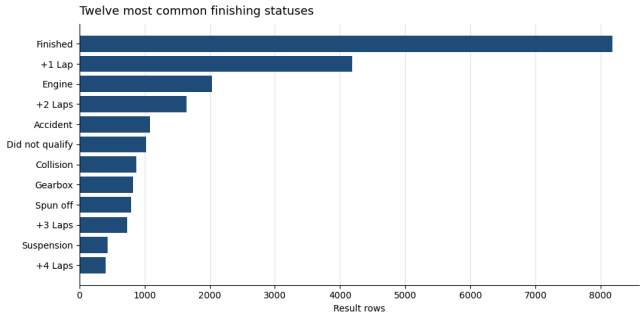


Figure 3. Most common finishing statuses.

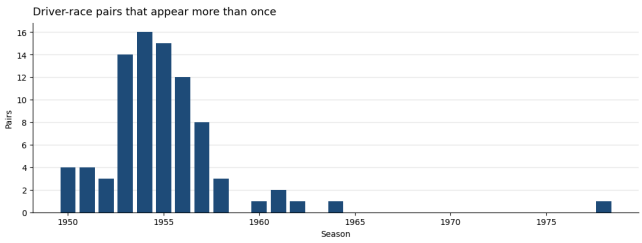


Figure 4. Duplicate driver-race pairs by season.

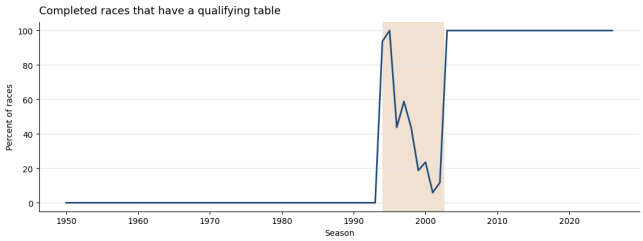


Figure 5. Races with qualifying data by season.

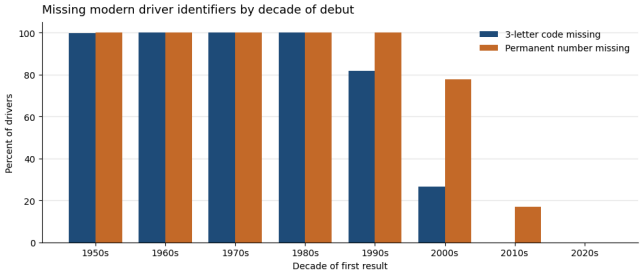


Figure 6. Missing driver codes and numbers by debut decade.

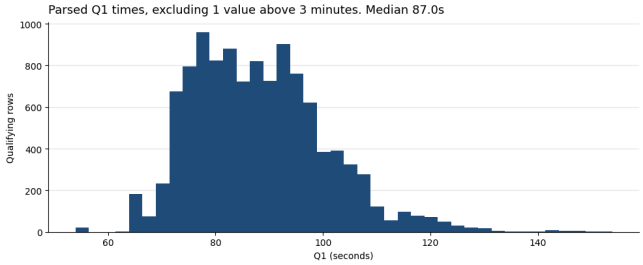


Figure 7. Parsed Q1 times in seconds.

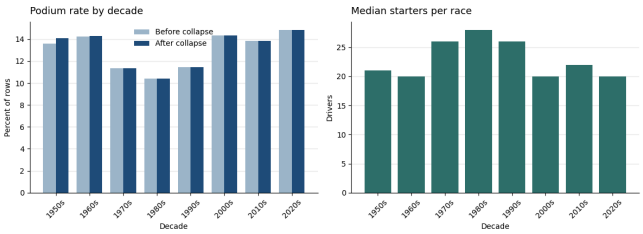


Figure 8. Podium rate by decade before and after the duplicate collapse.

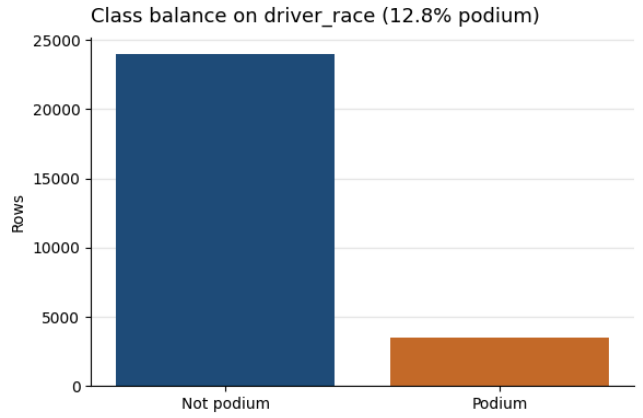


Figure 9. Class balance: 12.8% podium.

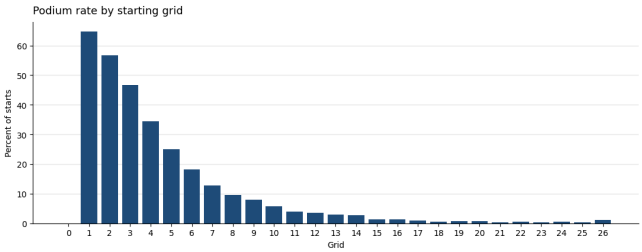


Figure 10. Podium rate by starting grid.

3. Data-engineering questions

Q1. Which circuits convert a front-row start into a podium most often, and does this change after 2014?

Method. driver_race → races → circuits, with qualifying for comparison. One row per start from grid 1 or 2; circuits need at least 20 such starts per era. The starting grid is used rather than the qualifying position, because penalties separate them: drivers who qualified on the front row and started there podiumed 72.6% of the time, against 28.9% for those moved back.

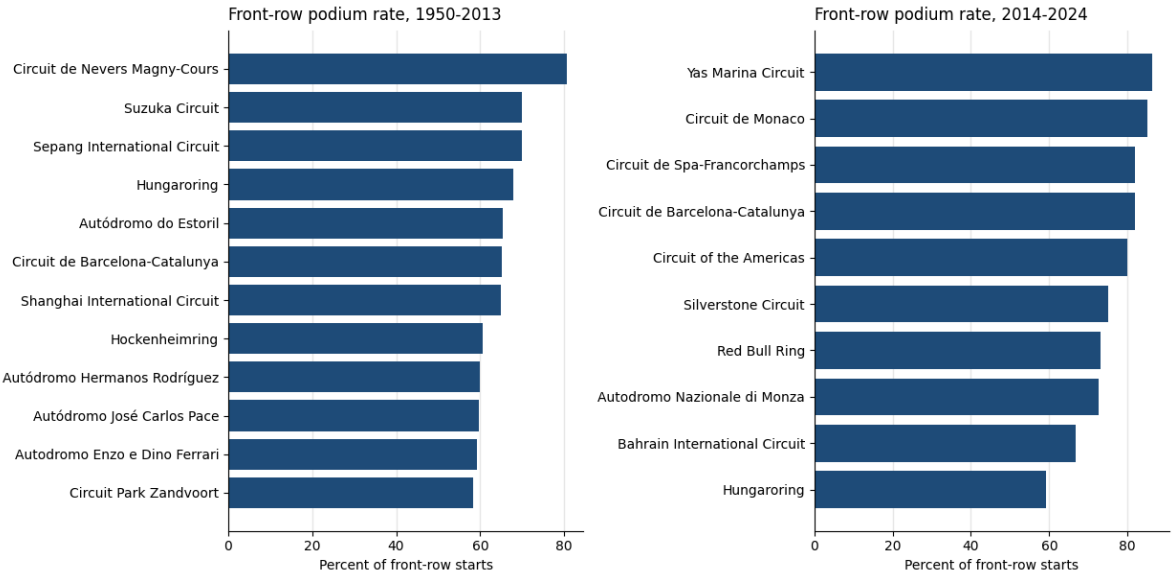


Figure 11. Front-row podium rate by circuit, 1950–2013 and 2014–2024.

1950–2013	Rate	2014–2024	Rate
Magny-Cours	80.6% (29/36)	Yas Marina	86.4% (19/22)
Sepang	70.0% (21/30)	Monaco	85.0% (17/20)
Suzuka	70.0% (35/50)	Barcelona	81.8% (18/22)
Hungaroring	67.9% (38/56)	Spa	81.8% (18/22)
Estoril	65.4% (17/26)	Circuit of the Americas	80.0% (16/20)

Answer. Yes, the ranking changes. Before 2014 Magny-Cours leads; from 2014 Yas Marina and Monaco lead. Monaco rises from 23rd to 2nd and Monza from 25th to 8th, while the Hungaroring drops from 4th to 10th. Conversion is also higher overall from 2014 (76% vs 68% in 2003–2013).

Q2. Do drivers podium more often at home, after controlling for starting position?

Method. driver_race → drivers (nationality) and races → circuits (country). Home and away rates are compared inside grid bands (pole, 2–3, 4–10, 11+). Rows with no grid slot are excluded.

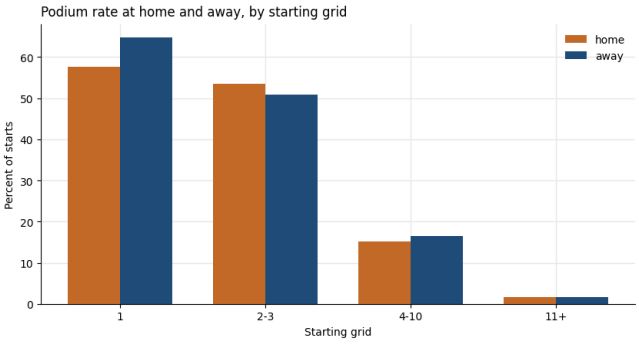


Figure 12. Podium rate at home and away by grid band.

Grid	Home	Away	Gap
Pole	57.6% (92)	64.7% (1,032)	−7.1
2–3	53.5% (172)	51.0% (2,074)	+2.5
4–10	15.1% (649)	16.6% (7,231)	−1.5
11+	1.6% (1,411)	1.7% (12,372)	0.0

Answer. No. Within each grid band the home rate is not higher; the weighted gap is −0.53 points and no band differs by more than 1.33 standard errors.

Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Method. driver_race → status → constructors. The denominator is race starts. A mechanical retirement is a status naming a failed component; crashes, driver errors, regulation decisions and the generic “Retired” are excluded.

Constructors need at least 30 starts per period.

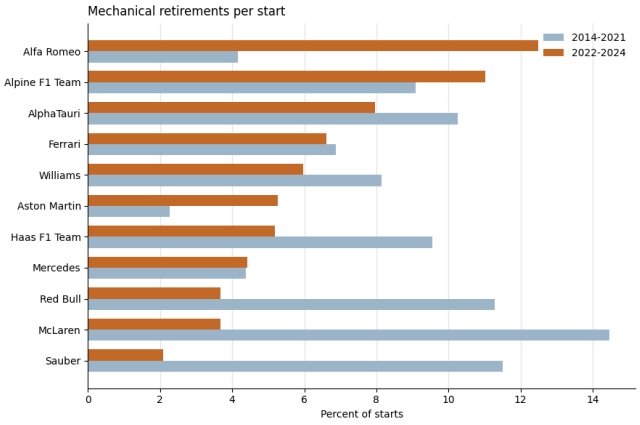


Figure 13. Mechanical retirements per start by constructor.

Constructor	2014–21	2022–24
McLaren	14.5%	3.7%
Sauber	11.5%	2.1%
Red Bull	11.3%	3.7%
AlphaTauri	10.3%	8.0%
Haas	9.5%	5.2%
Alpine	9.1%	11.0%
Williams	8.2%	6.0%
Ferrari	6.9%	6.6%
Mercedes	4.4%	4.4%
Alfa Romeo	4.2%	12.5%
Aston Martin	2.3%	5.3%

Answer. McLaren is highest in 2014–2021 (14.5%) and Alfa Romeo in 2022–2024 (12.5%). Across the field the rate falls from 10.1% to 6.1%.

4. Selected features

The 16 features fall into five groups, chosen from the training-year podium rates in Figure 14. Each is known before the race starts.

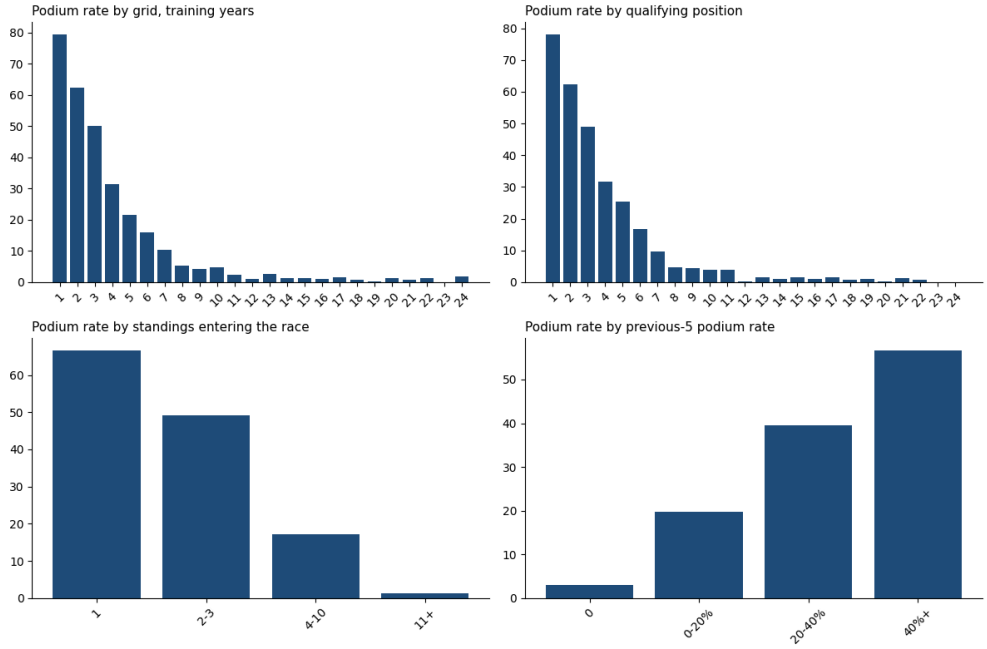


Figure 14. Training-year podium rate by grid, qualifying position, entering standing and recent form.

Feature	Source	Why it is known before the race
`grid`, `front_row`, `no_grid_slot`	results.grid	Starting grid is published before the start
`quali_position`, `reached_q3`, `quali_position_missing`	qualifying	Qualifying ends before the race
`driver_podium_rate_last5`, `driver_points_last5`	results, previous 5 starts	Earlier races only; current race excluded
`constructor_podium_rate_last5`	results, previous 5 team races	Earlier races only
`career_starts`, `form_missing`	results, earlier starts	Count of past starts
`driver_standing_entering`, `constructor_standing_entering`, `standing_missing`	standings at round − 1	Published after the previous round
`season_round`	races.round	Fixed by the calendar
`home_race`	drivers + circuits	Static facts

Excluded: the current race's result, points, laps, status, times, lap times, pit stops, standings and same-weekend sprint results.

5. Split and models

The split is chronological: train 2003–2018 (6,411 rows), validation 2019–2021 (1,200), test 2022–2024 (1,359). Hyperparameters and thresholds were chosen on validation only. The primary metric is PR-AUC because podiums are rare (about 15%); the secondary metric is F1 at the validation-chosen threshold.

Model	PR-AUC val / test	ROC-AUC val / test	F1 val / test	Precision val / test	Recall val / test
Grid rule (grid ≤ 3)	0.676 / 0.647	0.917 / 0.914	0.717 / 0.649	0.717 / 0.650	0.717 / 0.647
Logistic regression (baseline)	0.729 / 0.710	0.931 / 0.938	0.725 / 0.683	0.649 / 0.597	0.822 / 0.799
Random forest (selected)	**0.749 / 0.715**	0.937 / 0.938	0.733 / 0.683	0.665 / 0.613	0.817 / 0.770
Gradient boosting	0.721 / 0.684	0.936 / 0.913	0.729 / 0.635	0.719 / 0.658	0.739 / 0.613
Shallow NN (32, 16)	0.737 / 0.706	0.932 / 0.932	0.725 / 0.665	0.692 / 0.654	0.761 / 0.676

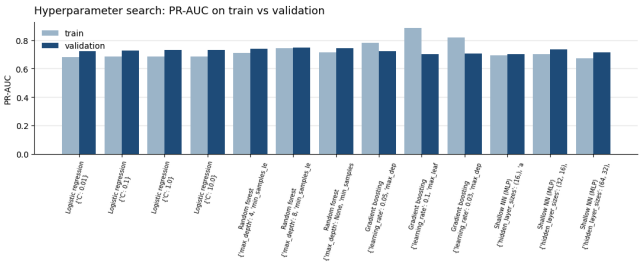


Figure 15. Train vs validation PR-AUC for every setting tried.

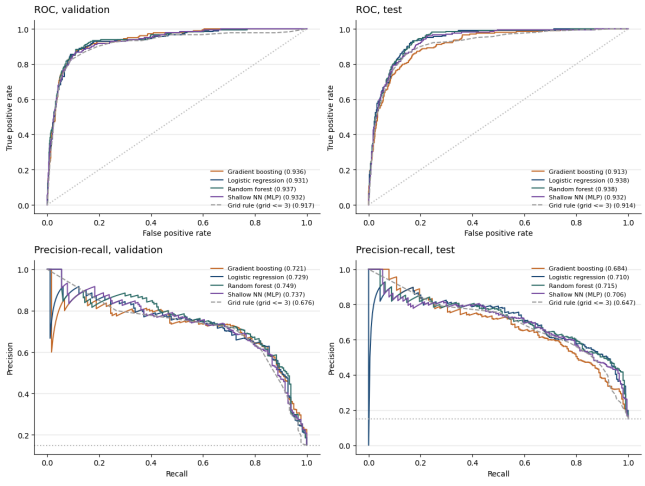


Figure 16. ROC and precision-recall curves.

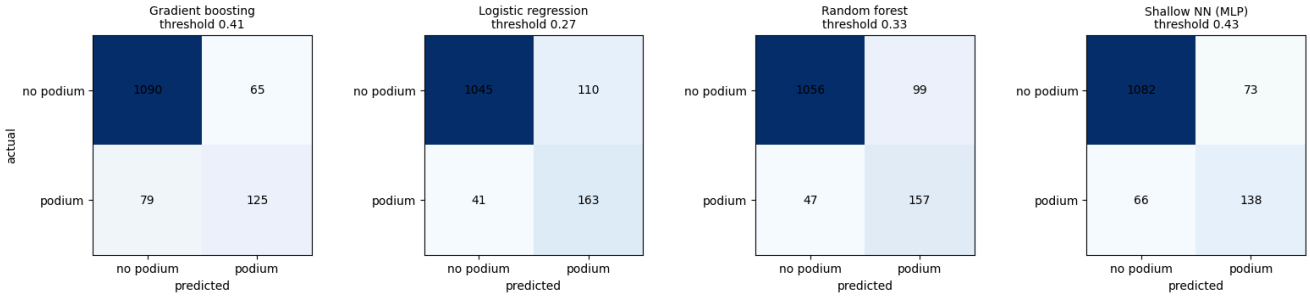


Figure 17. Test confusion matrices at the validation thresholds.

6. Ablation and preprocessing order

Removing one feature group at a time and refitting shows that grid matters most (validation PR-AUC drop 0.034), then qualifying (0.017) and standings (0.009). Form adds almost nothing (0.001) and home slightly hurts (−0.002). Swapping the order of clipping/quantile transform and imputation makes no difference for the tree model; scaling before or after imputation changes the logistic baseline by only 0.0002.

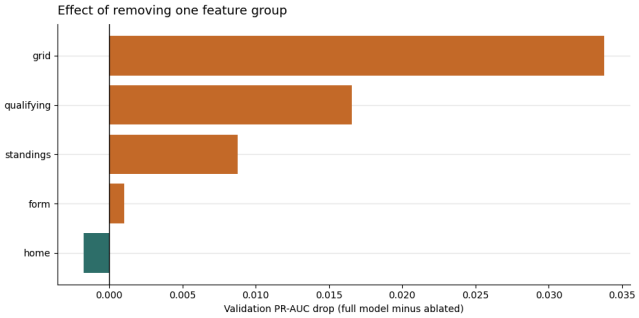


Figure 18. Validation PR-AUC drop per removed group.

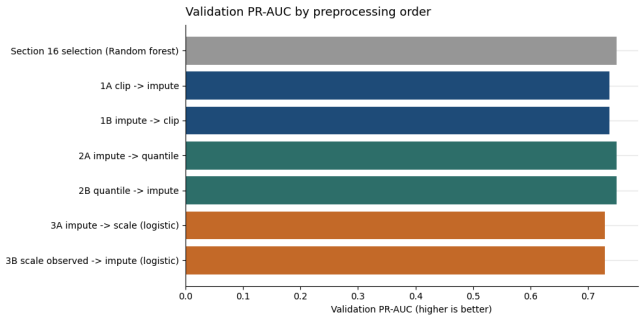


Figure 19. Validation PR-AUC by preprocessing order.

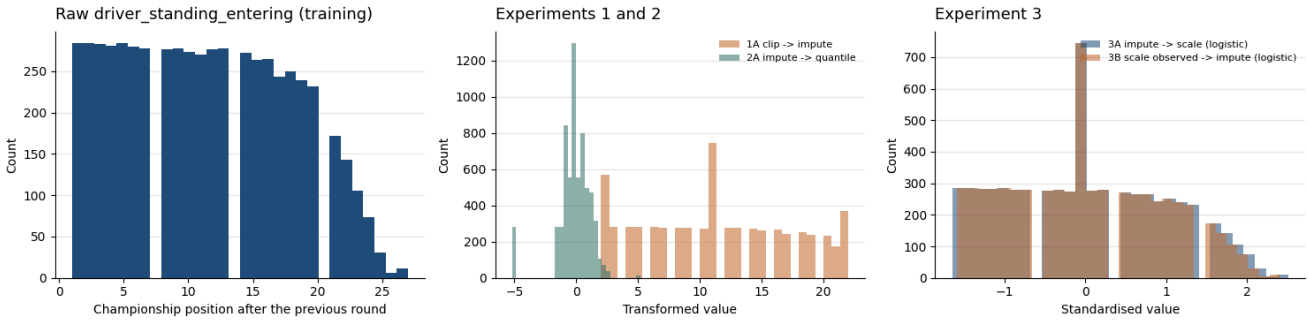


Figure 20. Effect of each preprocessing step on one feature.

7. Explainability

Permutation importance and SHAP agree that grid and qualifying position matter most, followed by recent form and the entering standing. For one validation row (Verstappen, 2021 Austrian GP, predicted 86.1%), SHAP and LIME both attribute the prediction mainly to pole position, P1 in qualifying and recent podiums. These explanations show what the model relies on; they are not causal effects.

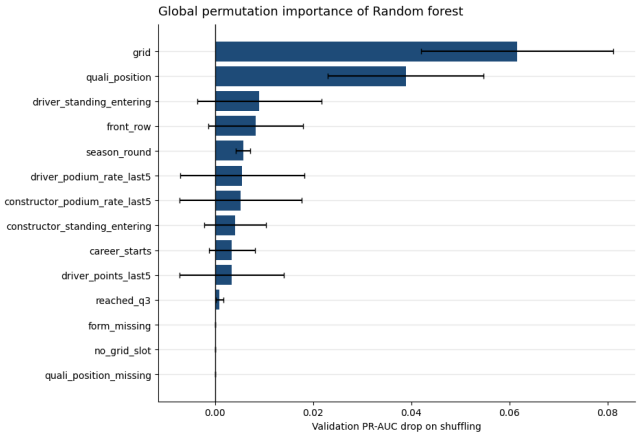


Figure 21. Permutation importance.

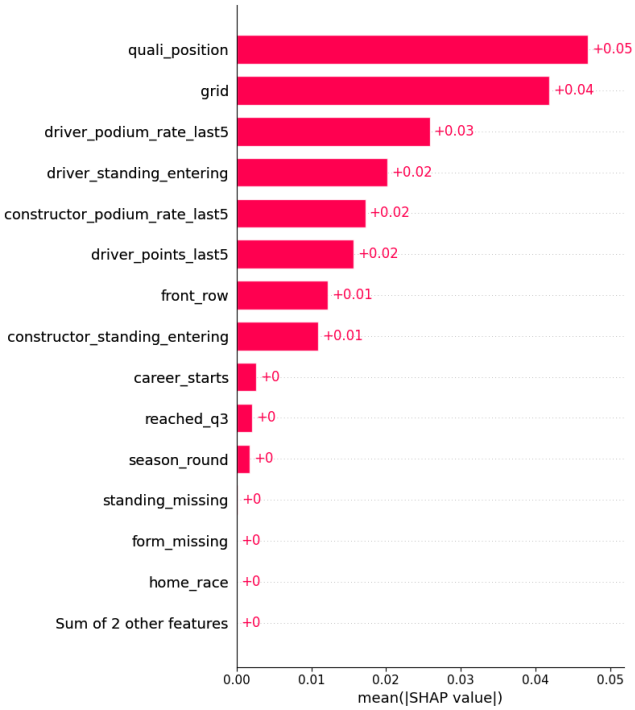


Figure 22. Mean |SHAP|.

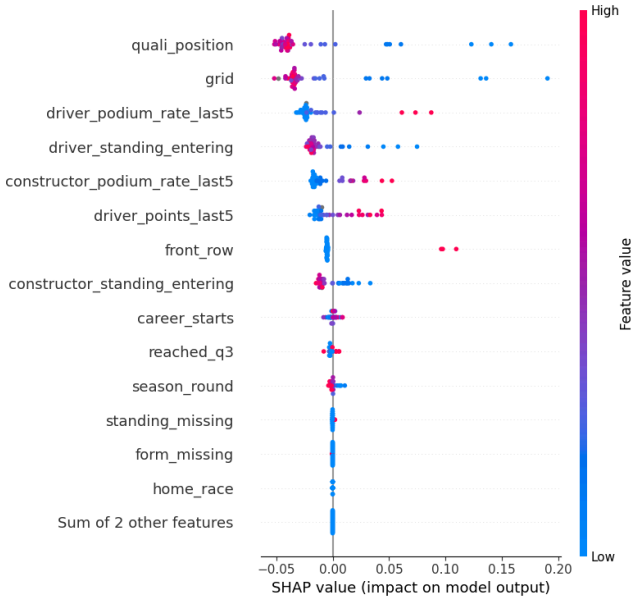


Figure 23. SHAP beeswarm.

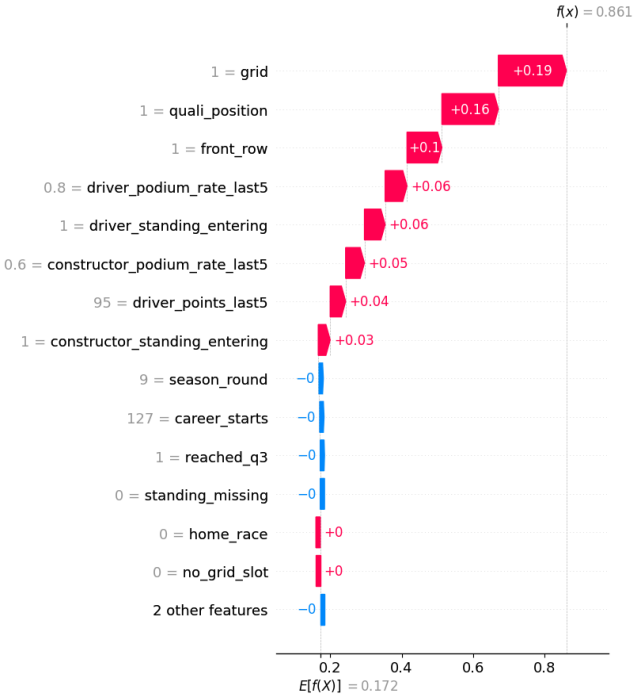


Figure 24. Local SHAP for one prediction.

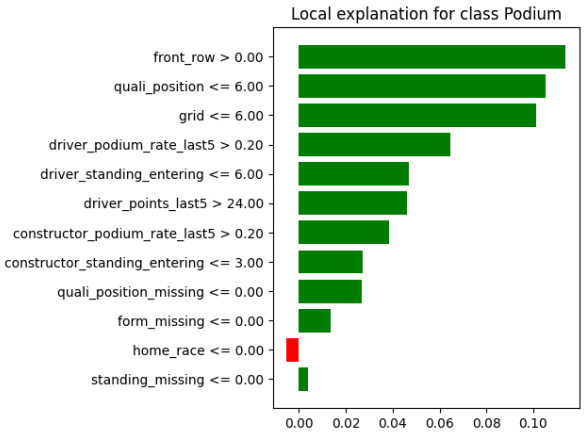


Figure 25. LIME for the same prediction.

8. Inference function

`predict_podium(record)` takes the pre-race fields for one driver-race, refuses post-race fields, and returns the podium probability, the decision at the validation threshold (0.333) and the missing fields. Missing fields are filled with training medians.

Record	Missing fields	P(podium)	Call	Actual
Complete (Verstappen, 2022 Canadian GP)	none	0.881	podium	podium
Partially missing (same entry)	qualifying and both standings	0.557	podium	podium

9. Known limitations

1. Qualifying data starts in 1994 and is complete only from 2003, so the model applies to 2003–2024 only.
2. Era sensitivity: the model was trained on 2003–2018; held-out PR-AUC varies from 0.670 to 0.779 by season (Figure 26), and 2022–2024 is a new rules era.
3. Missing inputs are rare in training, so a missing field is filled as a midfield value and lowers the prediction (0.881 → 0.557 above).
4. Only 17 pit-lane starts in training, so the model learned little from them.
5. Team renames reset constructor form (42 rows).
6. On 2021–2022 sprint weekends the grid differs from qualifying for 77–82% of drivers.
7. Q1 rankings rest on 20–26 front-row starts per circuit, so close ranks are within noise.
8. Q2 depends on matching nationality to circuit country.
9. Q3 excludes the generic “Retired” status, so mechanical rates are a lower bound.
10. Predicted probabilities are not calibrated, and a race can get more or fewer than three podium calls.

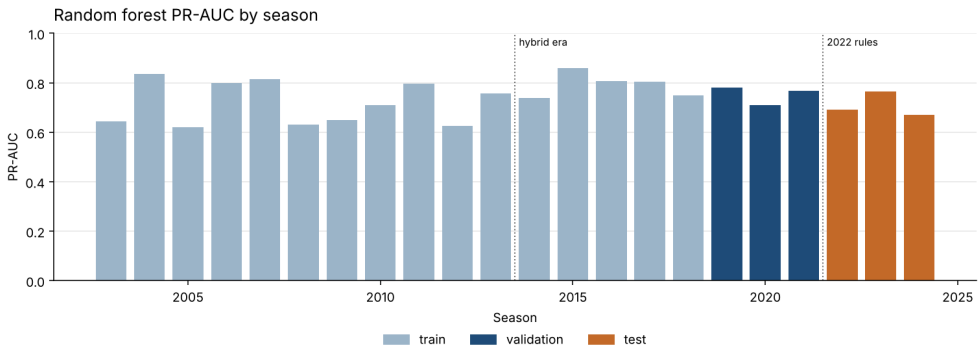


Figure 26. Random forest PR-AUC by season.